

FDR-Controlled KG Completion: Statistical Inference for Reliable Link Prediction

Step 1: Model & Scoring

Trained KG Embedding Models
TransE | RotatE | ComplEx | ConvE

Scoring function: $g(h, r, t) \rightarrow \text{score}$

Test set: $\{ (h_i, r_i, t_i^*) \}$

Step 2: p-value Construction

For each test triple:

Sample $K=100$ negative tails (filtered)

Compare true score vs. null distribution

$p_i = (\#\{s_{\text{neg}} \geq s_i^*\} + 1) / (K + 1)$

Output: m valid empirical p-values

Step 3a: BH FDR Control

Benjamini-Hochberg (BH): sort p-values, reject up to largest k where $p_{(k)} \leq (k/m) \cdot \alpha$

$R(\alpha)$ = number of statistically significant discoveries at guaranteed FDR level

Step 3b: q-value Estimation

Storey q-value: estimate π_0 (proportion of null hypotheses), compute per-hypothesis q-values

π_0 = fraction of test triples indistinguishable from noise | q_i = min FDR at which $H_0^{(i)}$ rejected

Step 3c: Empirical Bayes local FDR

Efron local FDR: transform $p \rightarrow z$ via inverse normal CDF, estimate mixture density $f(z)$

$\text{fdr}(z_i) = \pi_0 * \phi(z_i) / f(z_i)$ = posterior probability that triple i is a null (noise) discovery

Findings: $\pi_0 = 0.259\text{-}0.851$ (25-85% noise) | Per-relation FDR: 9.1%-96.5% (MRR-invisible) | MRR vs FDR: conflicting rankings (ConvE 2.8x TransE on BH power)